

Liu Zihan

Gender: Female Objective: New Graduate Backend Engineer Location: Beijing

Phone: 13000468352 Email: liuzihan1554@qq.com GitHub: <https://github.com/liuzihan660>

Education

2021.09 - 2025.06 Shandong University Communication Engineering (B.S.)

Courses: Operating Systems, Distributed Systems, Software Engineering, Data Structures and Algorithms, Computer Organization, Design Patterns

Summary

New CS graduate with two backend internships, solid Java/Spring Boot and database fundamentals and full project experience.

Internship Experience

2024.06 - 2024.12 NIO Intern Backend Engineer

- Rebuilt the config and template management service to support hot template reload without restart, cutting release time by 18%.
- Investigated rising instance memory rss using GC logs, heap dump and arthas, found an unbounded cache and cut memory by 41%, reused across multiple business lines.
- Led a production incident review with root-cause analysis and drove an alert-tiering and on-call rotation mechanism, and it became a reusable team best practice.

2023.07 - 2023.09 Beike Intern Backend Engineer

- Designed a Kafka-based async transaction pipeline with idempotency keys, retry queues and dead-letter handling, sustaining 8000 QPS at peak, reused across multiple business lines.
- Served 2120K requests per day while keeping core-path SLA above 99.9%, reused across multiple business lines.
- Fixed a log-collection failure caused by agent backpressure, restoring stability after tuning batch and buffer settings, improving end-to-end delivery quality.

Projects

Distributed Payment & Settlement Platform (Spring Boot + MySQL + Kafka + Redis)

- Used sharding and read-write splitting to keep single-table rows within tens of millions.
- Implemented daily reconciliation across payment gateway, ledger and settlement files with auto-correction, and it became a reusable team best practice.
- Recorded an audit trace id for every payment state transition for compliance, with experience documented and shared internally.
- Applied idempotency and eventual consistency to avoid duplicate charges on retry.

High-Concurrency Flash-Sale & Inventory System (Java + Redis + RocketMQ)

- Pre-deducted inventory in Redis with async DB writes to absorb traffic spikes.
- Designed distributed locks and token-bucket rate limiting to prevent oversell, with experience documented and shared internally.
- Stayed stable at 8000 QPS in load tests with error rate below 0.1%, recognized by stakeholders and peer

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- Built circuit breaking and fallback for downstream failures.

Enterprise Auth & Gateway Platform (Spring Cloud Gateway + OAuth2)

- Integrated OAuth2 and RBAC with multi-tenant isolation.
- Added circuit breaking to isolate faulty downstreams, recognized by stakeholders and peers.
- Unified auth, rate limiting, gray routing and tracing as reusable platform capabilities, supporting rapid business growth.
- Shipped a unified SDK to lower integration cost across teams.

Highlights

- Owned core modules that steadily served 2120K daily visits with 99.9%+ availability
- Continuously optimized MySQL indexing and slow-query tuning, cutting key-path latency from 1800ms to 320ms
- Led a major technical effort involving cross-team communication, staying stable at 8000 QPS peak
- Built a reusable methodology and docs around root-cause analysis and postmortems, adopted by several teams
- Drove a focused effort on CI/CD pipelines, reducing related issues by about 18%

Skills

Core skills: MySQL, Computer Networks, Docker and containerization, MySQL indexing and slow-query tuning

Proficient in: JVM tuning and troubleshooting, Redis, Java, Spring Boot

Familiar with: Distributed transactions and consistency, Kafka/RocketMQ messaging, Data Structures, Redis caching and distributed locks, Git

Self Evaluation

- Stays calm under pressure and recovers services quickly during high-concurrency incidents
- Strong communication and collaboration with product, QA and ops to deliver projects on time